

Engineering Capstone Specification: The-Raptors-Call V2.0 (Hybrid Mesh & Telemetry Defense)

1. Executive Overview & Humanitarian Mission

The-Raptors-Call V2.0 is an open-source, ultra-low-cost safety ecosystem designed to mitigate isolation-targeted violence, covert surveillance, and unauthorized community infiltration. The system integrates decentralized LoRa mesh networks, high-decibel acoustic deterrents, and localized mmWave radar. It provides a seamless, stress-optimized mobile React interface for first-responders and civilians, while a web-based Admin Command Center provides centralized vetting, telemetry, and threat triage.

2. System Architecture Diagram (Conceptual)

- Bottom Layer: Static LoRa Meshtastic Nodes (Tripwire) & Personal "Caller" Tags.
- Bridge Layer (Hardware/Software): Raspberry Pi / Local Machine running a Python WebSocket Bridge (reads Meshtastic USB serial packets and translates them to WebSocket broadcast).
- Compute Layer:
 - Supabase (Backend): Hosts PostgreSQL for data logging, Auth (via GUID JWT), and administrative ban lists.
 - Decentralized Cloud: Serves the React Frontend and stores "Rewind" forensic path history.
- Client Layer:
 - Mobile Browser: React Dashboard (user side).
 - Desktop Browser: React Admin Dashboard (vetting & command center).

3. Hardware Bill of Materials (BOM)

Loadout Hardware Key Components Est. Cost (USD)

1. Personal "Caller" Wearable TPU Sphere Seeded Studio XIAO ESP32S3 LoRa (915MHz), 130dB Piezo Buzzer, Magnetic Reed Pull-Pin, 400mAh LiPo + Qi coil. ~\$12.55
2. Smartphone Node Mobile Device Runs the React web dashboard. Provides localized GPS/Gyroscope/Thermal data. User Provided
3. Radar Dongle Smartphone USB-C Attach mmWave Radar (e.g., LD2410 or 24GHz) integrated into a slim USB-C module. ~\$15.00
4. Static Mesh Relays Outdoor Enclosures 4x LilyGo T-Beam / Heltec V3 devices with solar panel power management. ~\$35.00 each

4. Network & Telemetry Stack

4.1 LoRa Mesh Layer (Low-Bandwidth)

Standard Meshtastic 915MHz RF. Encrypted text packets and sensor triggers (motion/acoustic) relayed across a wide area independent of cellular towers.

4.2 High-Altitude PtP Backbone

Arboreal (canopy-mounted) relays utilizing excess solar power to drive high-bandwidth Point-to-Point (PtP) Wi-Fi bridges. This tunnels the slow LoRa data instantly back to the central server, bypassing multi-hop network latency.

4.3 The Python WebSocket Bridge (NEW)

- The Problem: The React dashboard cannot read a physical USB serial port directly from a web browser.
- The Solution: A lightweight Python script runs on a local Raspberry Pi or desktop machine.
 1. It uses pyserial to read the Meshtastic USB node.
 2. It opens a WebSocket server (using websockets or socket.io).
 3. When an emergency LoRa packet arrives, the Python script parses it and broadcasts a JSON packet over the WebSocket to all connected React dashboards in real-time.
- Implementation Note: This decouples the hardware from the cloud, enabling multiple admins to view the live feed simultaneously without proprietary closed-source bridges.

5. Software Architecture (Frontend & Backend)

5.1 Frontend (React SPA)

- Framework: React + Vite, utilizing HashRouter (to avoid 404s on deployment).
- Map Engine: react-leaflet with OpenStreetMap Tiles. User marker rotates via phone compass/gyroscope.
- State Management: React useState/useEffect handles local map states, overlay toggles, and emergency alarms.
- Admin Section (/#/admin): Renders AdminVettingForm.jsx and EmergencyDashboard.jsx (Handling the trilateration math from lib/trilateration.js).

5.2 Supabase Backend (New)

- Database: PostgreSQL (hosted on Supabase) for querying historical "Rewind" data and vetting records.
- Authentication via GUID: Email/password is disabled.
 - Workflow: The admin meets the client in person (or via encrypted video). The admin generates a GUID in the backend, hands the GUID to the client on paper/encrypted text. The client enters the GUID into the app to generate a JWT (JSON Web Token).
 - Benefits: No PII (Personally Identifiable Information) is ever stored in the database, ensuring immediate GDPR/CCPA compliance.

5.3 The "Silent Ban" Protocol (New)

Instead of deleting a banned user, use a Revocation List (Soft-Ban).

- Schema: Create a bans table (guid, mesh_node_id, banned_at, reason_internal).
- Logic: On every API request (guid/mesh_node_id check), the backend checks against the bans table. If found, it returns an HTTP 403 with a generic "Access Denied. Configuration mismatch."
- Why: The hostile user cannot tell if they are banned or if their hardware is broken, preventing them from re-accounting immediately. Additionally, because Meshtastic Node IDs are derived

from the hardware MAC address, if they try to generate a new GUID with the same radio, the ban list silently rejects the new vetting request.

6. Core Logic & Algorithms

6.1 RSSI/TDOA Trilateration (trilateration.js)

- Method: When 3+ static node relays hear a packet, lib/trilateration.js computes the Received Signal Strength Indicator (RSSI) and Time Difference of Arrival (TDOA).
- Result: Estimates an (X, Y) coordinate point on the Leaflet map. Plotted as a transparent bubble to visualize "Range of uncertainty".

6.2 The "Rewind" Auto-Forensic Algorithm

- 1. Continuous Logging: All environmental triggers (LoRa hits, motion, thermal) are logged in Supabase with timestamps and coordinates.
- 2. User Initiation: An admin clicks a Pink (At-Risk) or Red (Hostile) node and selects Review.
- 3. Pathfinding Query: The system queries the database, calculating backward in time from the entity's current location. It draws a localized spline curve connecting the triggered node path (e.g., Node C at 03:45 -> Node B at 03:20 -> Node A at 02:15).
- 4. Choke Point Intersection: The algorithm traces back until it hits a satellite/backbone camera at a perimeter road. It surfaces the exact video timestamp, revealing the entry point and stripping the anonymous threat of its mystery.

6.3 Radar Visualization Strategy (SVG Sweep vs. Advanced Plot)

- Primary Interface: SVG Sweep Simulation. A rotating radar cone (pie-slice graphic) animates over the map. When the mmWave dongle returns a distance and angle, a glowing dot pulses on the sweep.
 - Why: Under severe stress, humans cannot parse raw, disorganized FFT scatter plots. The sweeping radar mimics a familiar movie aesthetic—instantly recognizable as a tool providing "situational awareness." It is lightweight, runs 60fps on mobile, and builds user trust.
- Advanced Toggle: A small [Advanced View] toggle in the corner can switch to the raw 2D point-cloud for power-users verifying hardware data.

7. Presentation & Simulation Suite (UI Overlays)

A built-in Simulation Nav that auto-triggers modes during investor/community presentations, removing the need to fumble with real hardware during a pitch. These map directly to your React state overlays:

- Simulation A: The Vulnerable Person. Triggers a Pink node on the map. (Mimics bystanders flagging a confused/at-risk person).
- Simulation B: Hostile Infiltration. Triggers a Red node and instantly fires the "Rewind" spline to show their path back to the main road/choke point.
- Simulation C: Loud vs. Silent Alarm. Instantly toggles the bottom toolbar into the Amber (Silent Beacon) or Red (Sound Screamer) state.
- Simulation D: Proximity Chat. Instantly pops the chat window with a generated message: "Person X is approaching from the trail. Do you know them? Are you expecting company?"

8. Licensing Strategy (Standardized)

- Hardware: CERN Open Hardware Licence v2 (CERN-OHL-W-2.0) - Allows commercial use with reciprocity.
- Software/Firmware: GNU General Public License v3.0 (GPL-3.0) - Ensures the core mesh software remains free and open source.
- Documentation: Creative Commons Zero v1.0 (CC0-1.0) - Public domain documentation for other engineers to contribute.
